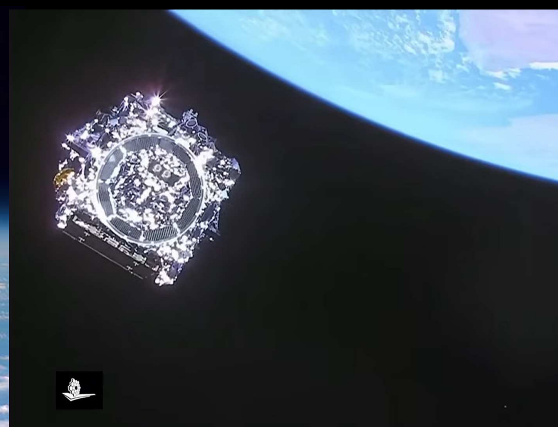
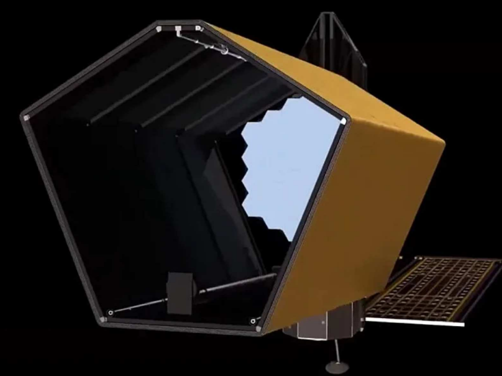
Lessons from JWST for future missions: HWO Perspective — What is required to make Mega-Science projects succeed?

Rogier Windhorst (ASU) — JWST Interdisciplinary Scientist

and Robert W. Smith (U. Alberta, Edmonton, Canada; HST& JWST historian)



Talk at the "JWST through Cluster Lenses" Conference, Aug. 12, Santander (Spain)

All presented materials are ITAR-cleared. These are my opinions only, not necessarily NASA's or ASU's.

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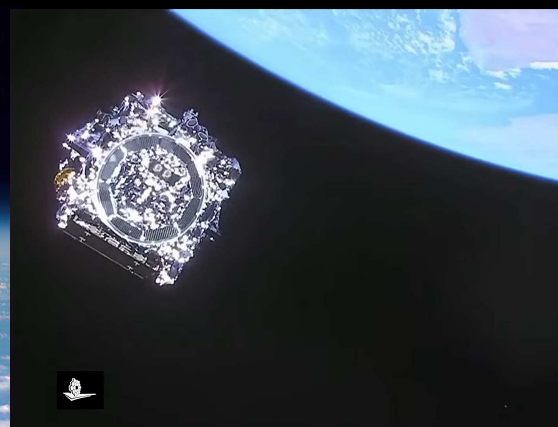

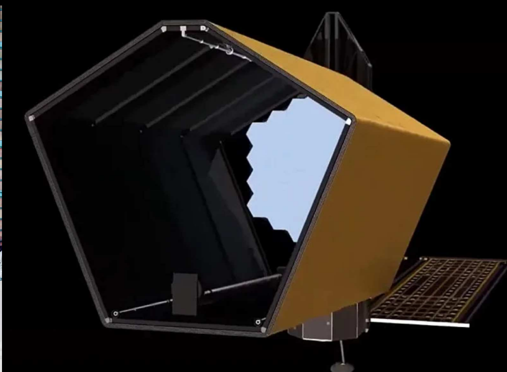

1969–2034^{+}(?)	1989–2046^{+}?	2010–2037?	2011–2070^{+}?
Launch: 1990	2021	Aug. 2026 (!)	$\gtrsim$2040?
Σ_{FC}: $\gtrsim$20 B$	$\gtrsim$10 B$	$\sim$3 B$	15–20 B$??

- My goal today: Inspire the younger folks in the audience to successfully build the Habitable Worlds Observatory (HWO).

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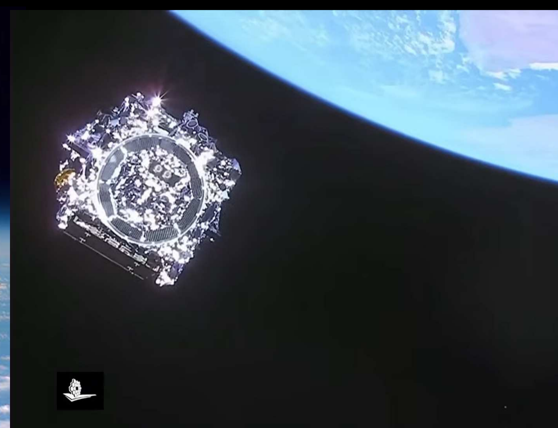
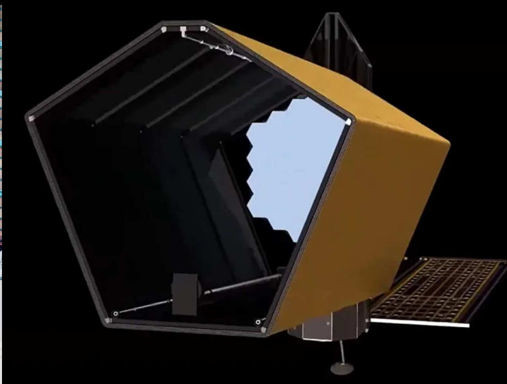
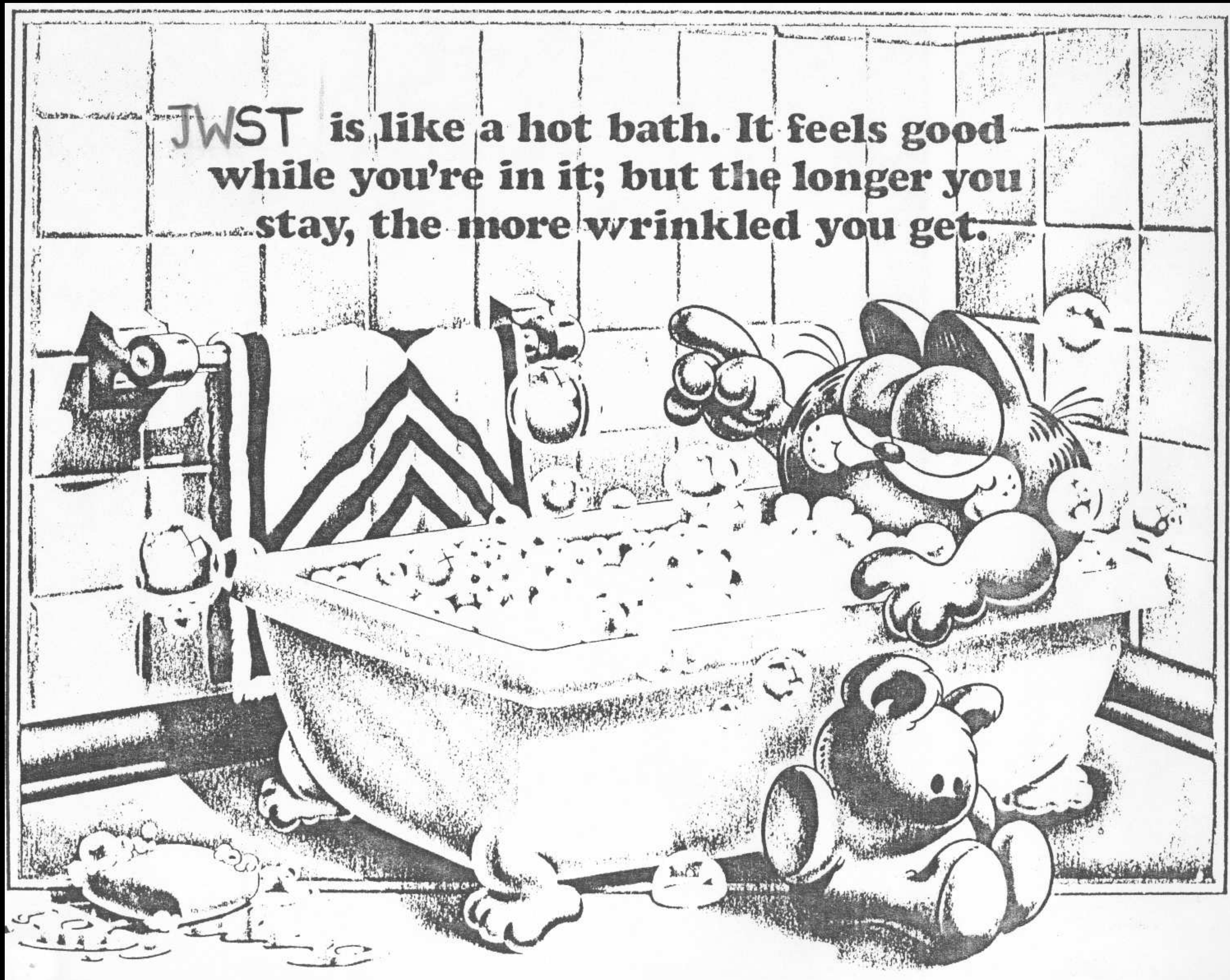




To those (understandably) concerned about events in the world today:

- HST survived 14 presidential, 28 congressional elections, 3 cancellations.
- JWST survived 9 presidential, 18 congressional elections, 2 cancellations.
- HST–HWO will span ∼25 US presidential & 50 congressional elections.

⟶ Maintain long-term vision to do space observing ∼30 years from now!



WARNING: Both Hubble and James Webb are $\gtrsim$40–50^{+} year projects:

You will feel wrinkled before you know it ... :) (chart from Jim Westphal's office, 1987).

Outline: Lessons from JWST

• JWST Lessons: Mega-project lessons also apply to HST & Supercollider. Key is that scale of efforts goes beyond what people are used to.

• Mega-projects demand new rules, in particular regarding building and keeping together a *strong Coalition* of project supporters and advocates.

Consumers Report: Very Good $\Rightarrow$ Good $\Rightarrow$ Neutral $\Rightarrow$ Fair $\Rightarrow$ Poor.

- (A) Scientific/Astro-Community Lessons
- (B) Technical Lessons
- (C) Management/Budget/Schedule Lessons
- (D) Political/Outreach Lessons

I thank Dr. Seth Cohen, Garth Illingworth, Rolf Jansen, John Mather, Eric Smith, Harley Thronson, & Ray Villard for comments.

Talk is on: `https://rogierwindhorst.github.io/windhorsttalks/venus26_jwstlessons.pdf`

Summary: Main Lessons from the JWST Project:

(1) Mega-projects demand new rules, in particular regarding building and keeping together a *strong Coalition* of project supporters and advocates:

(A) JWST Scientific/Astro-Community Lessons:

- 1) Project is a must-do scientifically and cannot be done any other way.
- 2) Keep advocating Mega-project to community until launch/first light.
- 3) Don't ignore importance of communication with patrons: Scientists, International partners, Contractors, Tax-payers, Congress, White House.
- 4) Don't have community infighting ("My mission is better than yours" — One key reason for Supercollider (SSC) demise).

(B) JWST Technical Lessons:

- 1) Use advanced technologies being developed elsewhere, if possible.
- 2) Know when not to select the most risky technologies.
- 3) Do your hardest technology development upfront. Have all critical components at TRL-6 before Mission Preliminary Design Review (PDR).

(C) JWST Management/Budget/Schedule Lessons:

• 1) Make conservative full end-to-end budget before Mission CDR.

• 2) Make sure budgets are externally reviewed, and at $\gtrsim$80% joint cost+ schedule confidence level. (Could not do $\lesssim$2010; Did so early 2011).

• 3) Plan & effectively use 25–30% ($+schedule!) contingency each FY.

(D) JWST Political/Outreach Lessons:

• 1) Assemble, maintain and fully use a broad Coalition of supporters and advocates who will fight for the project (SSC did so too late).

• 2) Have strong multi-partisan & multi-national support for project.

• 3) Strong technology benefits/lessons *TO* other parts of government.

• JWST *is* the telescope that the community asked for 1989-1996, and it was finally launched in 2021. Flagship missions have taken ∼16–32 years from first concept to launch!

OVERALL CONCLUSION: JWST got on the right track, but we did have to learn our programmatic lessons.

James Webb said in 1969: “In our pluralistic society, any major public undertaking requires for success a working consensus among diverse individuals, groups, and interests. A decision to do a large, complex job cannot simply be reached “at the top” ad then carried through. Only through an intricate process can a major undertaking be gotten under way, and only through a continuation of that process can it be kept going.”

(See quotes in front of Robert Smith’s 1993 book: “The Space Telescope”).

- Main message: Build and maintain a “Coalition” of strong project supporters for a “Mega-Project”.

- A Mega-project not only has to be technically feasible, but it also must be and remain politically feasible.

Edmund Burke (1729–1797): “Those who carry on great public plans must be proof against the worst delays, the most mortifying disappointment, the most shocking insult, and what is worst of all, the presumptuous judgment of the ignorant upon their design. ”

Avoid: “History *teaches* us that mankind *learned* nothing from history” (G. W. F. Hegel 1832).

(A) Scientific/Astro-community lessons from JWST

For a Mega-project to succeed, make sure that you DO:

- 1) Have ($\gtrsim$1) killer apps with full community support. (Be exciting enough that some dedicate most of their careers to make it happen).
- 2) Project is a must-do scientifically and cannot be done any other way.
- 3) Project highly ranked by community reviews/Decadal surveys.
- 4) Identify and highlight complementarity with other large facilities.
- 5) Still like the science and the project $\gtrsim$15–30 years later.
- 6) Offer project science and grant support to the whole community.
- 7) Keep *advocating* Mega-project to community until launch/first light.

(J. Bahcall and L. Spitzer said they had to "Sell" the HST Project to shepherd it through the approval process. We prefer to call it: "Advocate". We must make all stakeholders aware of mission purpose and progress throughout its long life cycle).

- Infighting killed the 1988 Superconducting Supercollider (SSC) in 1993 (Waxahachie, Texas; left).

- Canceled project funds never returns: Large Hadron Collider (LHC) and CERN didn't make that mistake (right).

$\Longrightarrow$ Avoid infighting with other (exoplanet) HWO stake-holders.

- Design HWO for exoplanets, reionization, and everything in between.

• Congress in 1990's: US can afford only one Mega-Science project at a time — this favored the International Space Station (ISS) over the SSC.

• Communication gap: Physicists had a "leave it to the experts" posture.

• Not clearly explaining the tangible economic, technological, and intellectual returns of high-energy physics to taxpayers weakened public support.

• Sagan: Physics community's elitist communication alienated the public.

• This dynamic contributed to resistance against later large-scale projects?

(A) Scientific/Astro-community lessons from JWST

For a Mega-project to succeed, make sure that you DON'T:

- 1) Have community infighting ("My mission is better than yours" — One key reason for Supercollider (SSC) demise).

(John Mather: "Management levels above the Mega-project need to help avoid community infighting, and work with advisory groups to ensure that everyone can see the choices. Complete openness is the key").

- 2) Have other projects canceled because of your Project, or perception thereof. Don't make enemies whenever possible.

- 3) Have science and grant support for a selected few.

- 4) Have GTO's be elite: they must serve & represent the community.

- 5) Ignore community input on project science priorities.

- 6) Ever ignore importance of great communication with U.S. patrons: Scientists, Contractors, Tax-payers, Congress, White House.

- 7) Ever ignore importance of great communication with foreign partners. (International projects are more robust politically, see e.g., SSC).

(B) Technical Lessons from the JWST Project

For a Mega-project to succeed, make sure that you DO:

- 1) Use advanced technologies being developed elsewhere, if possible.
- 2) Use latest proven technology where you can for killer science apps.
- 3) Know when not to select the most risky technologies.
- 4) Do your hardest technology development upfront. Have all critical components at TRL-6 before Mission Preliminary Design Review (PDR).

(Eric Smith: "Even after insuring TRL-6, you need to prove and test manufacturability of the technologies. Creation in a laboratory is not the same as proving something can be built reliably by industry.")

- 5) Only design to specs you need and can afford to fabricate & test.
- 6) Test, test, and retest where needed.
- 7) Have strong central control of systems engineering.

Any (space) mission is a balance between what science demands, what technology can do, and what budget & schedule allows ... (courtesy Prof. Richard Ellis).

(B) Technical Lessons from the JWST Project

For a Mega-project to succeed, make sure that you DON'T:

- 1) Use technologies below TRL-6 at Mission PDR.

- 2) Defer project Component PDR's or CDR's to well after Mission PDR or CDR, respectively.

- 3) Do system tests whose outcome do not make you change course.

- 4) Ask for 1μm diffraction limit unless you must have & can afford it.

- 5) [If you can't afford 1μm JWST diffraction limit, HOLD ground at 2.0μ, AND insist best effort made at 1μ without being cost-driver.]

- 6) Allow scientists to change requirements after Phase A (unless it is required to reduce risk).

Def: TRL-6 = "(Sub-)system model or prototyping demonstration in a relevant end-to-end environment (ground or space)."

Def: PDR = Preliminary, CDR = Critical Design Review.

(C) Management/Budget/Schedule Lessons from JWST

For a Mega-project to succeed, make sure that you DO:

- 1) Have competent *AND* project-friendly management in *ALL* of NASA.

- 2) Make conservative full end-to-end budget before Mission CDR.

(John Mather: Need correct cost analysis before final commitments are made. NASA has the responsibility to get it right).

- 3) Make sure budgets are externally reviewed, and at $\gtrsim$80% joint cost+ schedule confidence level. (Could not do $\lesssim$2010; Did so early 2011).

- 4) Plan *&* effectively use 25–30% ($+schedule!) contingency each FY.

(Harley Thronson: "Corollary of (3)–(4): Although cost and schedule control must be introduced from the start, do not (cannot) undertake costing too early in the process.")

- 5) Have a viable list of cost-saving and meaningful descopes.

(John Mather/Eric Smith: "Don't overestimate the benefit of descopes. After a commitment is made (in Phase C/D), it is extremely expensive — and may be an international incident — to delete components that have been accepted into the mission. Scientists must learn to accept descopes in Phase A — the power-point phase — when they are still feasible, and often essential to have a mission survive.")

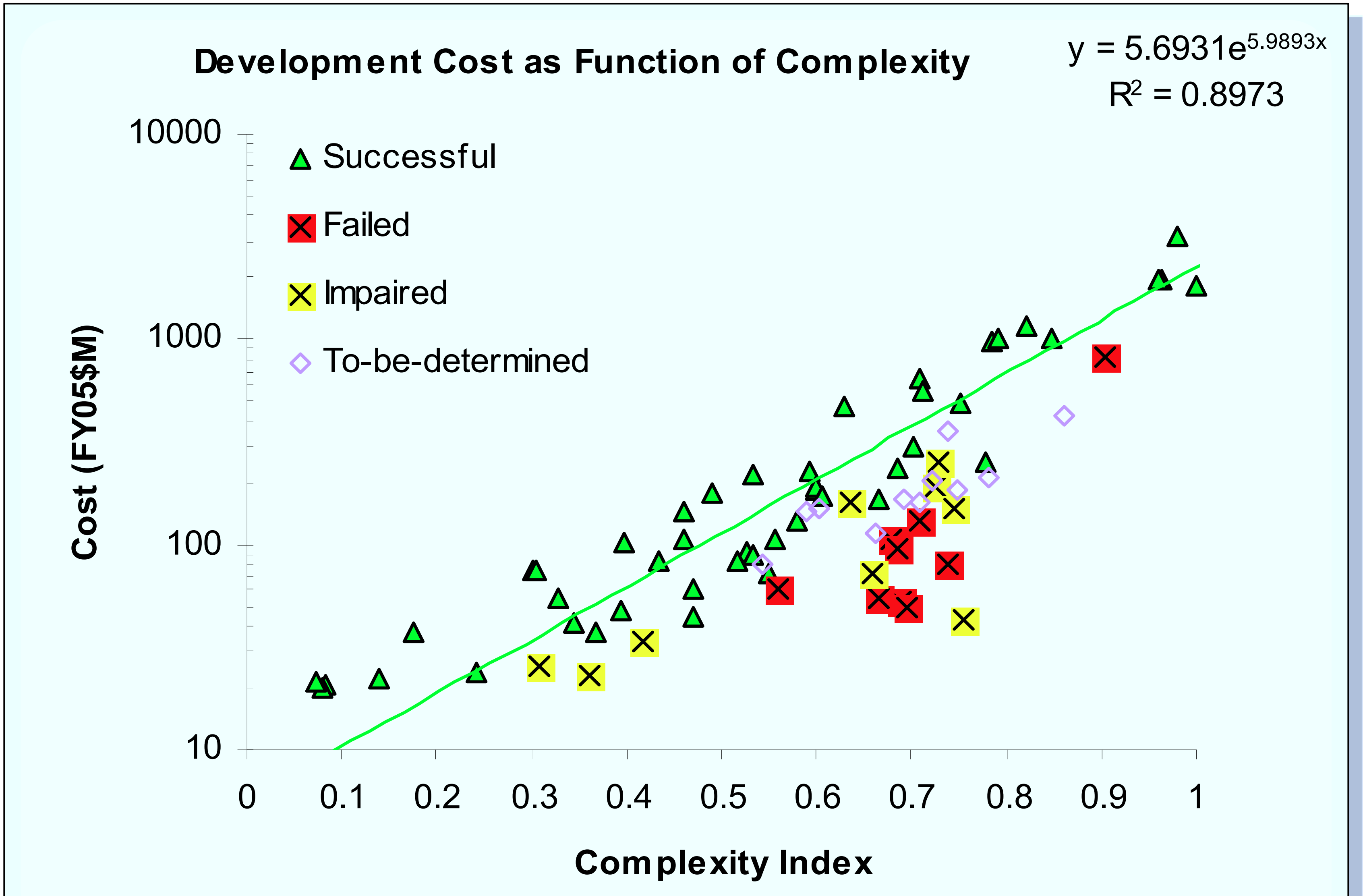
(C) Management/Budget/Schedule Lessons from JWST

For a Mega-project to succeed, make sure that you DO (cont):

- 6) Have great communication with all (sub-)contractors.
- 7) Put management pressure on contractors, when necessary.
- 8) Have best work-force from contractors for entire length of project.
- 9) Prioritize testing, and test extensively.
- 10) Carefully construct the incentives in your industrial contracts.

(Eric Smith: For-profit companies are very adept at insuring their teams win incentives. Hence, the Project contract needs to balance its incentives to get the best possible outcome in terms of both technical performance, cost *and* schedule).

When is a Mission Too Cheap?*



* "A Complexity-based Risk Assessment of Low-Cost Planetary Missions: When is a Mission Too Fast and Too Cheap?", Bearden, D. A.

(C) Management/Budget/Schedule Lessons from JWST

For a Mega-project to succeed, make sure that you DON'T:

- **1) Advocate project with (too) optimistic budget estimates.**

(Lesson number 1 from HST: Don't buy in at bargain prices):

HST was 450 M$ in FY78. At its 1990 launch, it was 1.5 B$ *without* the Shuttle launch or servicing cost. Jean Oliver said: "When someone asks you: Can you build it for xx $, the answer is invariably "Yes", and then folks go about making that work" (see Robert Smith's book).

(John Mather: "Our job includes communicating progress, and being as open with the world as possible. Eric Smith: Everyone is trying their best to understand the costs of things that have never before been attempted.)"

- **2) Cut project contingency to below critical mass (*i.e.* $\lesssim$25-30%/FY).**

When project contingency is cut to below $\lesssim$25-30%/FY, Project managers have no choice but to defer essential tasks to next FY's, which can ruin the Project's long-term budget plan very quickly.

- **3) Try (or allow Contractors to try) to save funds by cutting corners.**

(Eric Smith: If jobs are not properly allocated, the work can't be done for any price. You need individual key people and organizations, and strong central oversight.)

(C) Management/Budget/Schedule Lessons from JWST

For a Mega-project to succeed, make sure that you DON'T (cont):

- 4) Change science requirements after Phase A (unless essential to simplify, reduce risk and cost).

- 5) Allow contractors to change requirements, or have requirements jeopardize/delay project budget/schedule.

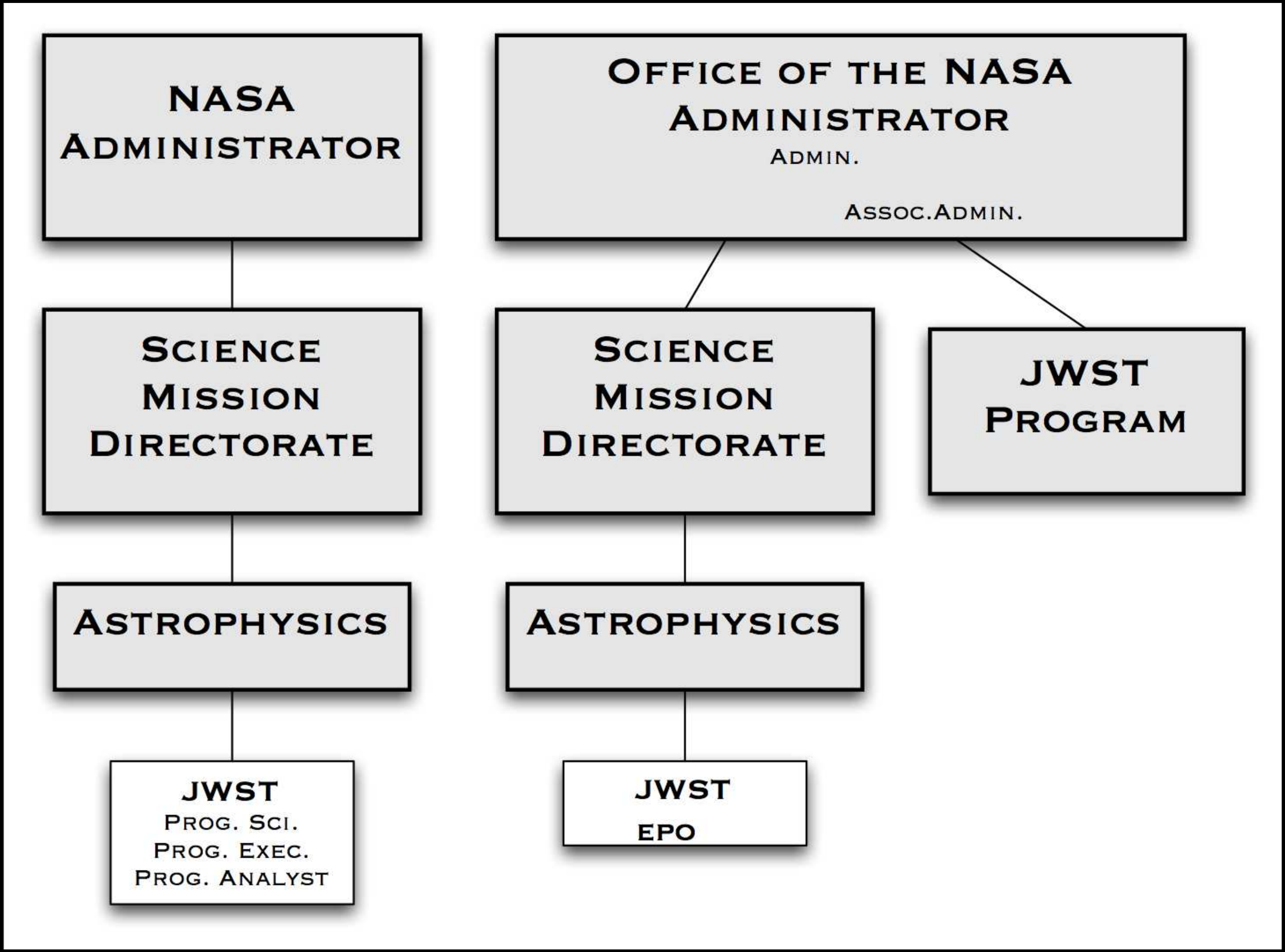
- 6) Change contract midstream, unless it is to reduce risk.

(Eric Smith: Sometimes it's is better to take the paperwork hit if it is technically required to change a requirement as you go).

- 7) Defer project Component PDR's or CDR's to well after the full Mission PDR or CDR, respectively.

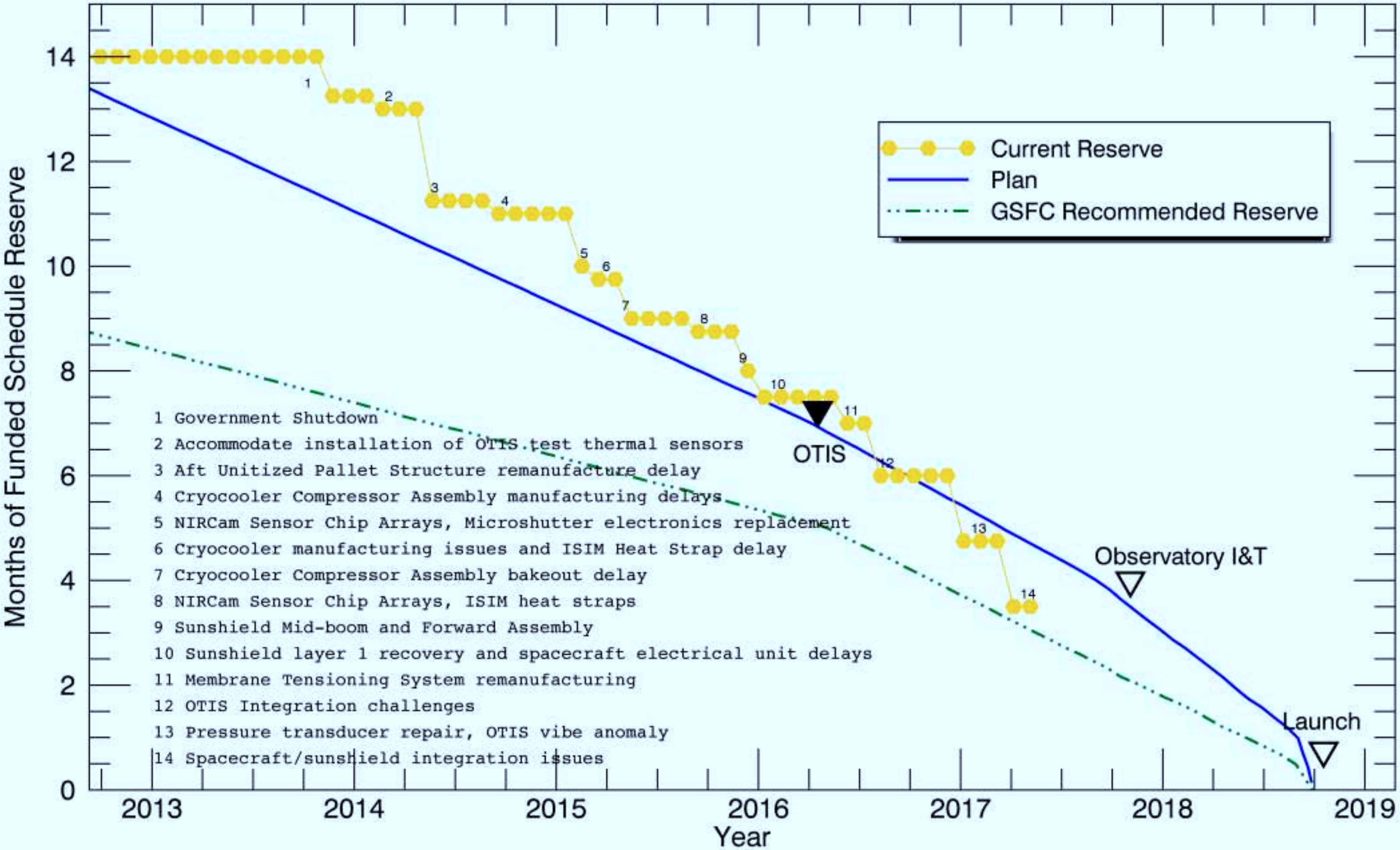
- 8) Test items without a clearly defined decision path.

How to launch JWST while minimizing impact on NASA Space Science?



NASA HQ Reorg: JWST budget no longer comes from SMD/ASD (Left); JWST was moved directly under the Administrator (Right; $\gtrsim$2011).

Funded Schedule Reserve



Keys to stay on schedule: 1) Sufficient Project contingency ($\gtrsim$25% of total).
2) Well replanned and managed Project (starting late summer 2011).

Mega-project needs heritage/links to technology from other parts of govt.

Mega-project needs strong technology benefits/lessons TO other parts of govt!

(D) Political/Outreach Lessons from JWST

For a Mega-project to succeed, make sure that you DO:

- 1) Assemble, maintain and fully use a *strong Coalition* of supporters and advocates who will fight for the project, since there will be storms and budget cancellations (HST did so successfully, SSC did so too late).

(Robert Smith: JWST didn't have its full "Coalition" in place until it got into serious budget trouble in 2010/2011).

- 2) Understand & foresee full political landscape of contractor world.

- 3) Have strong multi-partisan & multi-national support for project.

- 4) Educate, educate, and re-educate government & general public about project's essence.

(Harley Thronson: "Strong support for and opposition to Mega-projects can be as much *emotional* as rational. Not only does it matter to Coalition stakeholders whether it can be built, my state gets dollars, it is affordable, the technology is ready, and whether its science is exciting and possible. Both support and opposition can be emotional (or non-rational): pride, ego, bragging rights, reaction to animations and visualizations, and to project appearance/home institution, etc, all do matter. A Mega-project must be mindful of the very real human characteristics involved as well.")

(D) Political/Outreach Lessons from JWST

For a Mega-project to succeed, make sure that you DO (cont):

- 5) Strong heritage/links to technology *from* other parts of government.

(Eric Smith: "Unless an item is "build-to-print", the heritage benefit may not be great. We often have to modify items enough that we are making a new item effectively.")

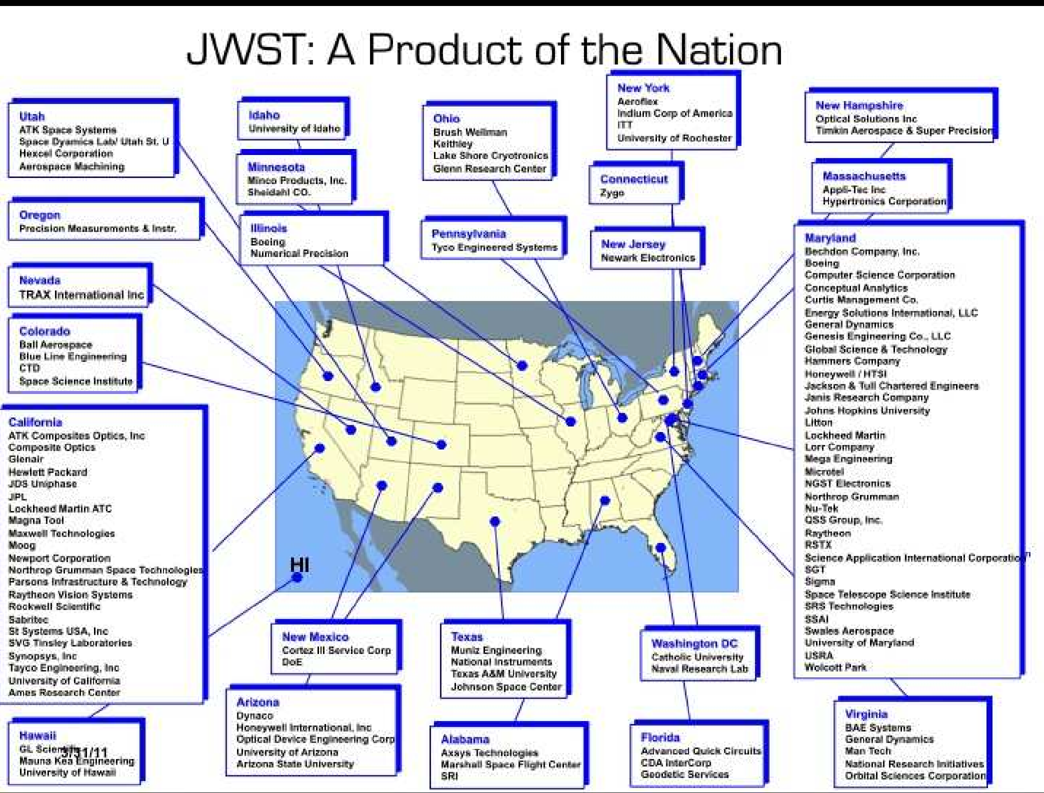
- 6) Strong technology benefits/lessons *TO* other parts of government.

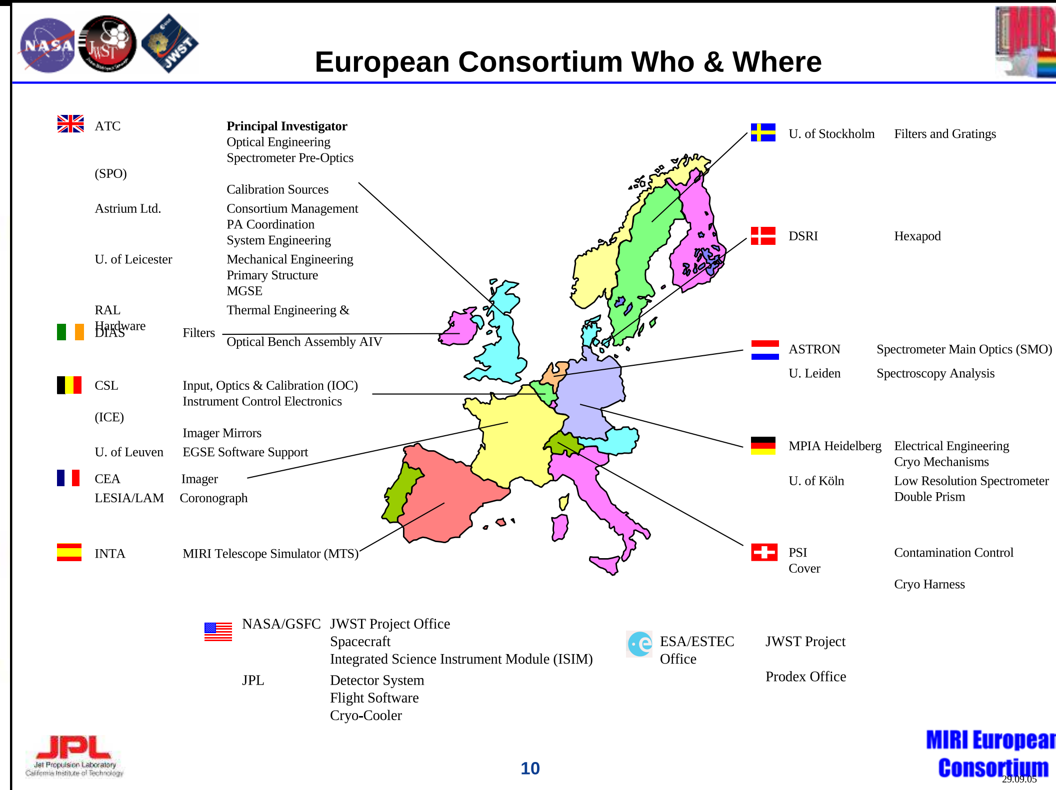
- 7) Strong, compelling benefits to society ("must-have" applications). (SSC could not explain to a broad audience: Why SSC?).

- 8) Know your (funding-) decision makers: what exactly makes them tick?

- 9) Have a last resort ("nuclear") option, but plan to never have to use it.

- 10) Expect the unexpected. This includes: be willing to do (legal and moral) things to help save a Mega-Project that may cost you your job.





- JWST hardware made in $\gtrsim$27 US States, Canada, and 16 European countries.

- Ariane V Launch & NIRSpec provided by ESA; & MIRI by ESA & JPL.

- JWST Fine Guider Sensor + NIRISS provided by Canadian Space Agency.

- JWST NIRCam made by UofA and Lockheed.

This nationwide + international coalition was critical for project survival!

Keep all levels of government informed about your mission:

Annual Girl Scout Stargazing at the White House South lawn (July 2015).

Amber Straughn (right; ASU graduate, now at NASA GSFC working for Dr. John Mather) informs the Obama's about NASA and JWST.

(D) Political/Outreach Lessons from JWST

For a Mega-project to succeed, make sure that you DON'T:

- 1) Have project politicized in the government (lesson from SSC).

(John Mather: Entire professional societies went to war to stop the Space Station. They didn't have to do that. This rubbed off negatively onto the SSC as well ... Astronomy has done well because our Decadal Survey sets priorities, which the community is willing to accept and abide by).

- 2) Assume your government understands or likes the project: Educate, educate, and re-educate.

- 3) Have project become target of social media: Must continuously educate instead and reach out to opponents.

- 4) Have project too concentrated in one state (or nation): MUST distribute efforts and wealth.

- 5) Don't pick fights you cannot win.

- 6) Ever fall asleep, not until launch anyway ...

A personal take on science and society

World view

By Rogier Windhorst

Save Hubble: the race to preserve the space telescope

The orbiting observatory's unique vision and public support continue to be crucial for understanding the Universe.

The Hubble Space Telescope is an audacious expression of human curiosity. Since its launch in 1990, it has done much more than just expand humanity's scientific reach: it has changed how people see themselves in the cosmos.

Now, Hubble's future hangs in the balance. A NASA working group is considering whether to extend the observatory's mission into the 2030s by boosting it into a higher, more stable orbit, or to send a robotic mission to decommission it, plunging the telescope into the ocean.

Each option's risks, benefits and costs need to be considered. Calls for white papers and pitches to define Hubble's science priorities for the next decade will conclude in July. Later this year, the working group will share its advice with NASA and the US Congress, which will inform further plans.

I submitted a white paper arguing that Hubble should remain operational for as long as technically feasible. It was an easy case to make, but the future of this iconic telescope is not guaranteed. Scientists and the public must back this remarkable project into its fifth decade.

What's at stake? Hubble's unique ability to complement all of NASA's current and planned scientific missions until 2040. The ultraviolet and optical wavelengths that the telescope captures are essential to understanding how and when galaxies, stars and planets formed.

Hubble has swept us beyond human imagination. It tore back the curtain on an exciting incandescent Universe – not just for a handful of astronomers, but for everyone. Its images are visceral, both beautiful and terrifying at the same time, and it revealed the raw interplay of matter and energy on scales that words could never fully express.

Its chain of discoveries reinvigorated modern astrophysics. The deepest optical image of the Universe ever taken revealed how galaxies have evolved throughout cosmic history. Measurements of supernovae showed that the Universe's expansion is accelerating under the influence of the mysterious force known as dark energy. Hubble mapped the cycle of stellar birth and death with unprecedented clarity and was the first telescope to measure the atmosphere of a planet orbiting a star other than the Sun.

The mission prioritizes transparency, collaboration and public engagement. Its data have been used by astronomers around the world. Far from obsolete, its instruments are still cutting edge, operational and in demand – proposals outnumber availability by a factor of six-to-one. And it fills a crucial niche in NASA's suite of flagship observatories.

Hubble's blue–UV view of the Universe complements that of the James Webb Space Telescope, which captures infrared light with comparable resolution and field of view. Broad coverage of the electromagnetic spectrum is essential for characterizing extrasolar planets, galaxies and black holes. A successor to Hubble, the Habitable Worlds Observatory, won't be launched before 2040.

Scientific progress depends on continuity as much as it does on innovation. And Hubble's legacy of long-term observations – a baseline of 36 years and counting – is also crucial for studies of black-hole variability, supernova environments, cosmic distances and changes in the atmospheres of planets in the Solar System.

The telescope's future relies on political will, public support, sustained funding and international cooperation. Its running costs, at tens of millions of dollars a year, sound expensive, but a new observatory would need billions. And investing in Hubble supports people, operations, grants and expertise, enhancing the returns of other missions.

Orbital maintenance raises urgent questions, however. Hubble's orbit is decaying slowly and might become unstable by 2035, when a peak in solar activity will swell Earth's atmosphere and increase drag on the telescope. A boost to a higher orbit, or a controlled end-of-life mission, would require an engineering study, risk assessment and funding.

NASA has explored privately operated boosting possibilities, and other options might emerge. Now, it must evaluate the possibilities transparently, consider their costs and risks, and describe the scientific implications of each. If NASA and the US Congress conclude, after careful study, that extending the mission is not the best path forwards, the scientific community and the public should understand the reasoning. If viable options exist to preserve Hubble's capabilities, those opportunities deserve serious consideration.

The public has a stake. Hubble is one of the rare scientific facilities with a cultural connection. Public support depends on emotionally resonant storytelling. Hubble images have inspired many young people to become engineers, scientists and teachers.

My call is direct. Keep Hubble's scientifically unique modes operating while NASA, the US Congress, the astronomy community and the public decide on its future. Preserve the grants and expertise that allow its data to shape science and to inspire and train the next generation. Commission and conduct a trade-off study that considers reboosting, servicing and end-of-life options. State clearly what 'uniquely Hubble' science would be lost if no action is taken.

Hubble reflects the paradox of big science: it is fragile yet resilient, vulnerable yet enduring. It requires constant tending – political, financial, cultural – and yet, when it works, it produces knowledge that outlasts the people who built it by generations.

"Hubble is one of the rare scientific facilities with a cultural connection."

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ARIZONA STATE UNIV.

CONCLUSIONS:

- Both HST & JWST were well worth fighting for!

- I spent most of my career doing that (see, *e.g.*, 2026, *Nature*, 655, 284),

- and I hope that I inspired many of you to do the same for HWO!

SPARE CHARTS

Some things are better left discussed during ...

Miller time!

Het Borrel uur!

La Hora Feliz!

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Fiscal Year 2017 JWST HQ Milestones

Month		Milestone	FY2016 Deferral	Comment
Oct-16	1	Complete portable clean room for Telescope and Science Instruments (OTIS)		Completed 10/13/16
	2	Complete final checkout of new shaker tables at Goddard Space Flight Center	•	Completed 10/13/16
	3	Begin making electrical connections between spacecraft panels		Completed 10/7/16
	4	Complete Sunshield Mid-Boom Assembly #2 functional test	•	Completed 12/5/16
Nov-16	5	Start optical measurements of OTIS prior to vibration and acoustic tests		Completed 10/24/16
	6	Deliver Science and Operations Center release 1		Completed 9/30/16
	7	Perform Cryocooler installation into the spacecraft bus and begin functional testing		Completed 10/29/16
	8	Complete Aft Unitized Pallet Structure assembly	•	Completed 10/29/16
	9	Deliver Aft Unitized Pallet Structure to Observatory I&T	•	Completed 3/14/17
Dec-16	10	Deliver Forward Sunshield Pallet Structure to Observatory Integration and Test (I&T)	•	Completed 3/28/17
	11	Start OTIS vibration and acoustic testing program		Completed 11/19/16
	12	Complete final test of engineering model of telescope center section at Johnson Space Center (JSC)		Completed 10/31/16
	13	Deliver sunshield flight membranes to Observatory I&T		Completed 12/15/16
Jan-17	14	Complete OTIS vibration and acoustics testing		Completed 3/2/17
	15	Deliver observing proposal and planning subsystem software build that supports launch		Completed 1/12/17
	16	Complete electrical testing of the spacecraft at Northrop-Grumman		Completed 3/7/17
Feb-17	17	Complete OTIS optical measurements after vibration and acoustic tests		Completed 3/31/17
	18	Deliver wavefront and control software that supports launch (controls telescope mirror shape)		Completed 1/20/17
	19	Deliver horizontal deployable radiators to Observatory I&T		Delayed June for release testing
Mar-17	20	Deliver OTIS to the Johnson Space Center		Completed 5/7/17
	21	Deliver the pre-launch Flight Operations System software build		Completed 2/17/17
	22	Delivery of sunshield extension boom #2 membrane attachment assembly to Observatory I&T		Completed 4/13/17

Blue font(underline) denotes milestones accomplished ahead of schedule, orange font denotes milestones accomplished late. "•" denotes 2016 milestones carried forward.

Milestones: How the Project reports its progress monthly to Congress.

Key is constant & full communication, and always see problems coming.

Milestone Performance

- Since the September 2011 replan JWST reports high-level milestones monthly to numerous stakeholders

	Total Milestones	Total Milestones Completed	Number Completed Early	Number Completed Late	Deferred to Next Year	Deferred more than one quarter
FY2011	21	21	6	3	0	0
FY2012	37	34	16	2	3	3
FY2013	41	38	20	5	3	2
FY2014❖	36	23	10	8	11	10
FY2015	48	44	22	12	4	3
FY2016	45	39	25	7	6	2
FY2017	38	24	11	17*	3	1

*Late milestones have been completed late within the year or are forecast to complete late within the year. Deferred milestones are not included in the number-completed-late tally.

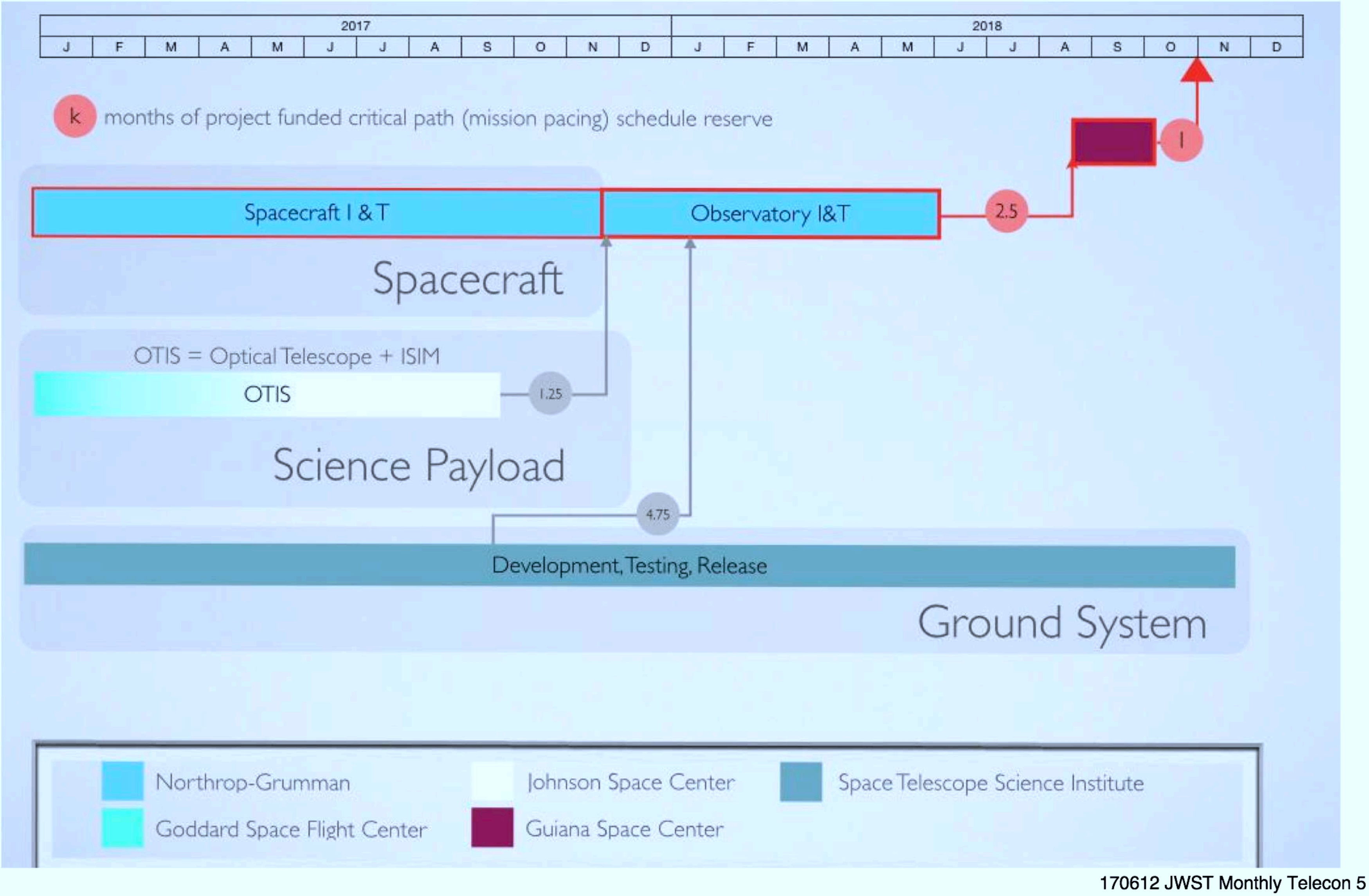
❖ Milestone accounting in FY2014 was complicated by the government shutdown and multicomponent milestones

FY14: 8 milestones late by 1 month due to Oct 13 Government shutdown.

FY15, F16: Most "Lates" not on critical path, nor cause a launch delay.

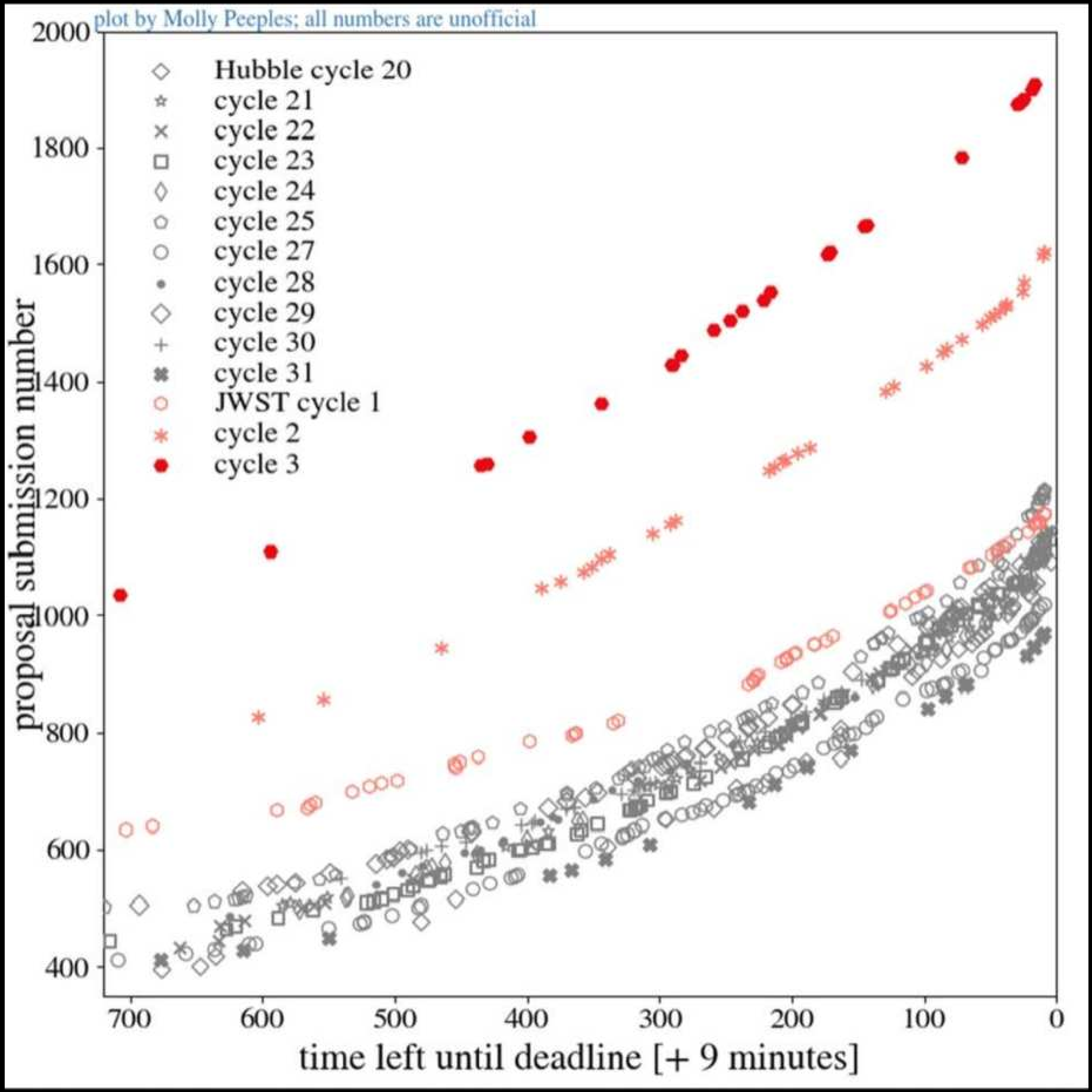
FY17: Most "Lates" anticipated to finish with FY, not causing launch delay.

Simplified Schedule



Path forward to Launch (in Oct. 2021): $\lesssim$10 months schedule reserve.

Instruments+detectors & Optical Telescope Element remain on critical path.



• Webb is now THE highest-in-demand NASA Flagship mission ever,
but Hubble remains in at least as high a demand as it was 30 years ago!

(1) SCIENCE IMPACT BY THE HST & JWST COMMUNITY (Feb. 2025):

- HST: $\gtrsim$500–1000 refereed papers/year by the community since 1990.
- $\gtrsim$50,600 HST papers on ADS, 1,050,000 citations since 1990, h_{HST}=339!
- JWST: $\gtrsim$4700 refereed papers, h$\simeq$155 (144k cites) since July 2022!
- In year 1–5: JWST already outdoing HST's yearly production.

(2) NEWS RELEASES BY THE HST & JWST COMMUNITY (Feb 2025):

- NASA's Hubble Space Telescope (HST) had $\gtrsim$1250 science press releases since 1990, each with $\sim$400 million readers (or impressions) worldwide.
- $\sim 500\times10^9$ reads (or impressions) of Hubble press releases in total $\Rightarrow$
- *On average* each human on Earth would have read $\gtrsim$60 Hubble stories during their lifetimes.
- HST is the most publicized space astrophysics mission in NASA history.
- JWST: $\gtrsim$170 press releases since 2022, each 0.5–1 billion readers.
- JWST is now the most-in-demand space mission in NASA history.
- ASU Cosmology: 10 billion readers from $\gtrsim$10 releases since 2022 (URL).

The "Impressions" metric: industry standard for news exposure ("impressions") can appear squishy when values exceed the world population.

- Yet they are standard at Meltwater, Critical Mention, LexisNexis, etc.

What Impressions Actually Measure: how often readers have the opportunity to encounter the same story across multiple outlets and platforms.

- Reflect repeated exposure across screens, newspapers, and syndication — not unique readers.

Head counts are not available: proprietary to online news organizations:

- Monitoring services are not permitted to access them!
- Algorithms differ across services, so impression totals vary and cannot be corrected to a true head count.

Using Impressions: For press purposes, relative exposure is what matters.

- Typical Hubble stories: 100 million–800 million impressions.
- Webb stories: frequently billions, reflecting "new kid on the block".

Actual attentive audience: Science-literacy researcher Jon Miller: ∼35 million Americans attentive to NASA's space exploration efforts.

- This figure has remained essentially steady for three decades.

DISCLAIMERS:

- (1) The materials below are our opinions only (Rogier Windhorst & Robert Smith's), not necessarily NASA's or our Universities' opinion.

- (2) When we get to *(D) Political Lessons* and suggest "One should", or "One should not", we do NOT imply to address civil servants. (I.e., only those individuals who are allowed to reach out to Congress should consider doing so, or should ever feel encouraged doing so).

- (3) No NASA funds or resources are used to reach out to Congress, nor are University funds or resources used to reach out to State legislators.

- (4) No ITAR sensitive materials are ever presented in outreach talks or activities for Mega-Projects, and certainly not in such talks given abroad.

- (5) We are not here to judge, but to learn.